

CODEHIVE

JAVA PROGRAMMING: DEFINITE ITERATION (FOR LOOPS)

► What is a For Loop?

The **for loop** is used when you know exactly how many times the code should run. It is more compact and organized for counters than a while loop.

► Modern Loop Syntax

```
for (init; condition; increment) {  
    // Statement 1: Runs once at start  
    // Statement 2: Checked before every iteration  
    // Statement 3: Runs after every iteration  
}
```

► Practical Examples

Printing 0 to 4:

```
for (int i = 0; i < 5; i++) {  
    System.out.println(i);  
}
```

Multiplication Table (Factorial logic):

```
int n = 5; int fact = 1;  
for (int i = 1; i <= n; i++) {  
    fact *= i;  
}
```

► Enhanced: The For-Each Loop

Best for iterating through **Arrays** or **Collections** without using an index counter.

```
String[] cars = {"Volvo", "BMW", "Ford"};  
for (String car : cars) {  
    System.out.println(car);  
}
```

► Control Keywords

- **Break:** Exit the loop immediately.
- **Continue:** Skip the current iteration and move to Statement 3.

```
if (i == 2) continue; // Skips the rest of code for i=2
```

► For Loop Summary

- **Definite Count:** Always preferred when the iteration count is known.
- **Nesting:** Inner loops run completely for every single outer loop tick.
- **Counter scope:** Variables declared in the loop header (like `int i`) are local to that loop.
- **Infinite Loop:** `for ( ; ; )` creates an intentional infinite loop.